Tab 1

Vivekanand Education Society's Institute of Technology

Department of Computer Engineering

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Project Synopsis Template (2025-26) - Sem VII

NutriVision: AI-Powered Food Recognition & Calorie Estimation

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Abstract

This project introduces an AI-powered food intake and diet monitoring system that combines multi-modal input processing with modern artificial intelligence capabilities. The system enables users to upload food images or record voice descriptions through a web interface, while AI models perform food classification, speech-to-text conversion, and natural language processing to extract food items and quantities. Nutritional values, including calories and macronutrients, are retrieved from the USDA FoodData Central database. It incorporates computer vision, speech recognition, and NLP principles, offering a scalable and intelligent solution for personalized dietary tracking and health monitoring.

Introduction

NutriVision is an AI-powered system for monitoring dietary intake using images and voice inputs. It uses computer vision to classify food from images, speech recognition to convert voice descriptions to text, and NLP to extract food items and quantities. Nutritional data is fetched from the USDA FoodData Central database to provide accurate calorie and macronutrient estimates, offering a quick and user-friendly way to track daily food consumption.

Problem Statement

Accurately tracking daily food intake is a challenge for many individuals due to the time and effort required in manual logging and estimating nutritional values. Existing diet tracking methods often rely on user-entered data, which can lead to inaccuracies caused by memory lapses, portion misjudgment, or lack of nutritional knowledge. There is a need for an intelligent, automated solution that can identify food items from images or voice descriptions and provide precise nutritional information instantly.

Proposed Solution

NutriVision offers an AI-powered, multi-modal approach to diet monitoring by allowing users to either upload a food image or record a voice description of their meal. For image inputs, computer vision techniques preprocess and classify the food item using a CNN model trained on the Food-101 dataset. For audio inputs, a speech recognition model converts the voice to text, and NLP algorithms extract food names and quantities. The identified food items are then matched with nutritional information from the USDA FoodData Central database, and values are adjusted based on portion sizes. Finally, the system displays calories and macronutrients in an easy-to-read format, enabling quick and accurate dietary tracking without manual data entry.

Methodology

The proposed system employs artificial intelligence techniques to accurately identify food items and estimate their nutritional content from both images and voice inputs. Food images from the Food-101 dataset are preprocessed and classified using a CNN with transfer learning. Audio inputs are processed through a speech-to-text model (Whisper) and NLP to extract food names and quantities. Recognized items are matched with the USDA FoodData Central database to retrieve nutritional values, which are adjusted for portion sizes. Results, including calories and macronutrients, are then displayed through a web-based interface for easy dietary tracking.

Block Diagram

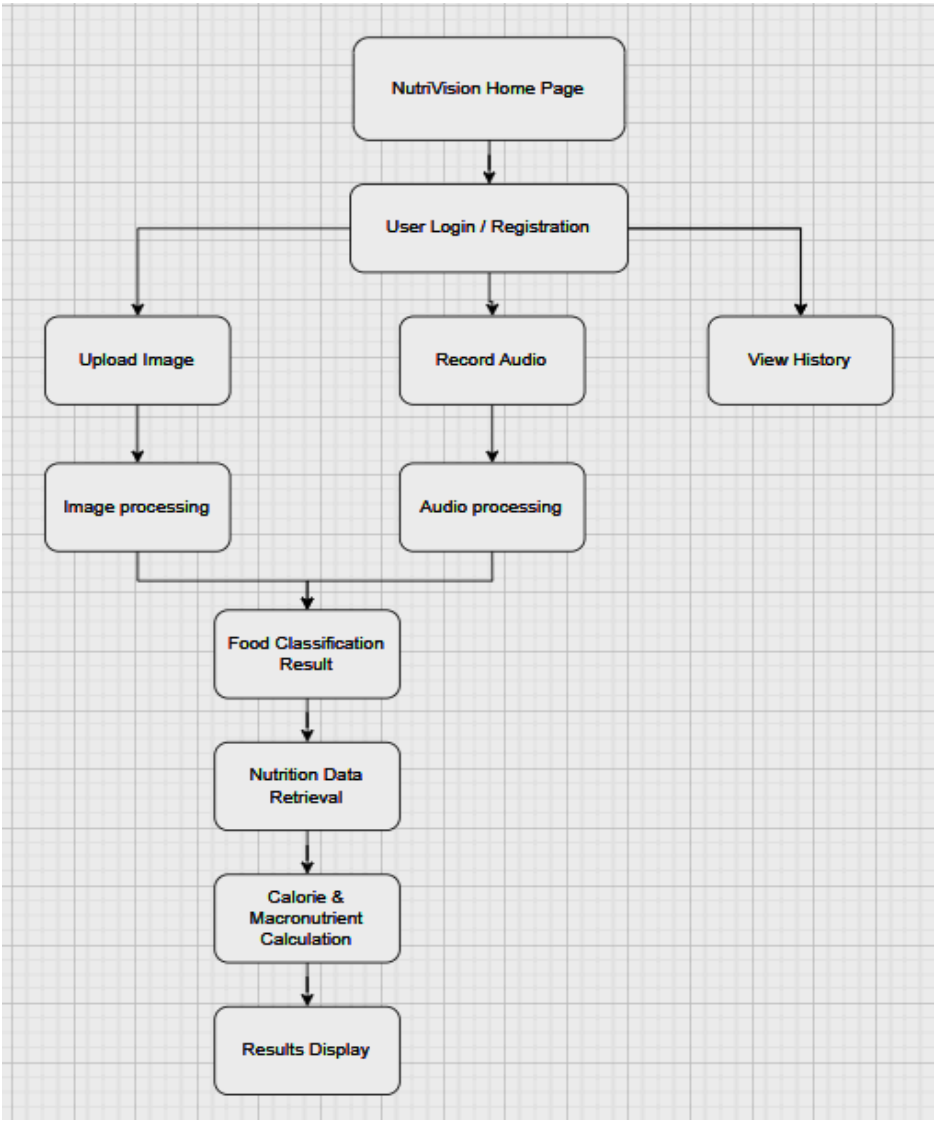

Fig.1.1 Block diagram of the website

Hardware Requirements

- **Processor:** Intel i5 or higher
- **RAM:** Minimum 8GB
- **Storage:** Minimum 100GB HDD/SSD
- Internet Connection

Software Requirements

- **OS:** Windows 10 / Linux / macOS
- **Backend:** Node.js / Django / Express
- **Frontend:** React / Angular / HTML/CSS
- **Database:** MongoDB / MySQL

Tools and Technologies

- **React.js & Tailwind CSS** – For building the web application interface.
- **Python (Flask/Django)** – Backend server and AI model integration.
- **TensorFlow/PyTorch** – Food image classification and calorie estimation.
- **OpenAI Whisper** – Speech-to-text for audio food input.
- **MySQL** – Database for storing user data and food logs.

Proposed Evaluation Measures

- **Image Classification Accuracy** – Percentage of correctly identified food items from uploaded images.
- **Speech-to-Text Accuracy** – Correct transcription rate of audio food descriptions.
- **Nutritional Data Accuracy** – Comparison of estimated values against verified nutrition databases (e.g., USDA).
- **System Response Time** – Average time taken from input submission to results display.
- **User Satisfaction** – Feedback ratings collected via surveys after system use.

Implementation

- **Dataset Preparation** – Collect Food-101 for image training and USDA FoodData for nutrition lookup. Map Food-101 labels to USDA entries.
- **Image Model Training** – Preprocess images (resize, normalize, augment) and train a CNN (e.g., EfficientNet or ResNet) for food classification.
- **Audio Processing** – Use Whisper for speech-to-text, then apply NLP to extract food names and quantities from transcriptions.

- **Backend Development** – Create API endpoints to handle image/audio uploads, run models, retrieve nutrition data, and return results.
- **Frontend Development** – Build web UI for login, uploading media, displaying nutrition results, and viewing history.
- **Database Integration** – Store user details, uploaded inputs, and nutrition logs for history tracking.
- **Evaluation & Testing** – Measure image classification accuracy, speech-to-text accuracy, nutrition lookup correctness, and system response time.
- **Deployment** – Host the application on a cloud platform, ensuring smooth model inference and API performance.

Gantt Chart

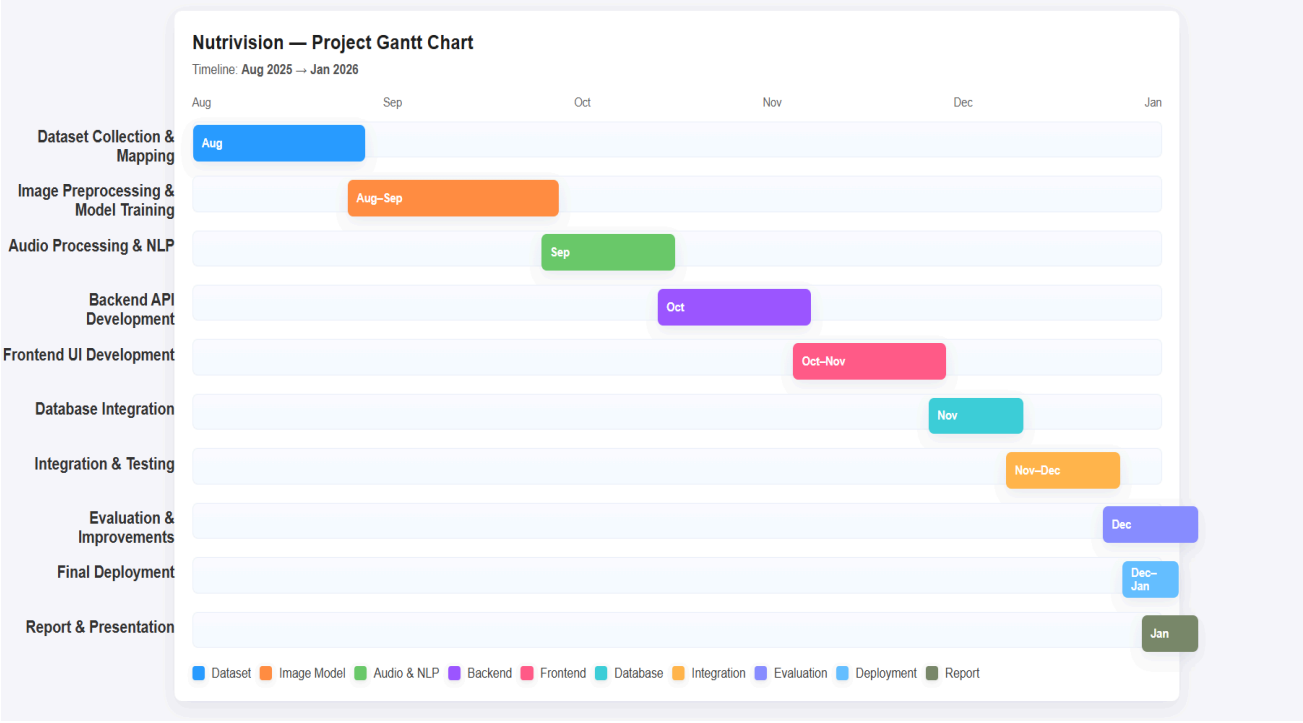

Fig 1.2 Gantt Chart

Comparative Study

Aspect	NutriVision	MyFitnessPal	Bite AI	CalorieMama
User Input Methods	Image upload + Voice description	Manual text entry + Barcode scan	Image upload only	Image upload + Manual entry
AI Technologies Used	Computer Vision (EfficientNet), Speech-to-Text (OpenAI Whisper), NLP	Basic database matching, no AI image/speech	Deep learning image classification only	Computer Vision + OCR for text recognition
Datasets for Training	Food-101, USDA FoodData Central	Proprietary food database	Custom food image datasets	Public food datasets + User data
Nutritional Info Accuracy	High (USDA standardized data)	Medium (user-entered or scanned data)	Medium-high (image recognition errors possible)	Medium (OCR errors + user corrections)
Deployment Model	Cloud backend (Flask/FastAPI + TorchServe), web + mobile ready	Mobile and web apps with cloud DB	Mobile app focused	Mobile + web hybrid
Personalization Features	Diet tracking + AI-driven recommendation s	Basic tracking + manual goal setting	Limited personalization	Tracking + some AI insights
Strengths	Multi-modal input, scalable AI pipeline, USDA integration	Large user base, barcode database	Strong image recognition accuracy	OCR integration for menus and labels
Limitations	Requires server setup, heavier AI	Manual input prone to errors	No voice input, image-only	OCR errors, partial

	compute			automation
Overall Recommendation	Best for AI-powered multi-modal dietary monitoring	Good for users preferring manual input & scanning	Best for fast image-based food recognition	Useful for label/menu reading with some automation

Conclusion

The NutriVision system successfully integrates AI-driven image recognition and speech processing to provide an efficient and user-friendly platform for food identification and nutritional analysis. By leveraging deep learning models for visual classification and natural language processing for voice commands, the system enables quick and accurate calorie estimation from multimodal inputs. This approach not only simplifies dietary tracking but also promotes healthier eating habits, offering significant potential for expansion into healthcare, fitness, and personalized nutrition domains.

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Chapter 3

Chapter 3: Requirements

3.1 Proposed Model

The proposed system, NutriVision, is an AI-powered nutrition tracking platform designed to automate food recognition and nutrient estimation using computer vision, natural language processing (NLP), and speech recognition. It allows users to log their meals by simply uploading images or using voice input.

The system identifies the food item, estimates portion size, and retrieves nutritional data (such as calories, proteins, fats, and carbohydrates) from FoodData Central, ensuring high accuracy and convenience. By integrating these intelligent modules, NutriVision minimizes manual logging errors, enhances user experience, and promotes healthy eating habits through data-driven insights.

The architecture comprises the following major components:
- Image Processing Module: Uses deep learning (CNN) models for food recognition from uploaded images.
- Speech Processing Module: Converts audio descriptions into text using OpenAI Whisper and extracts food details via NLP.
- Backend Module: Handles data flow, integrates AI models, retrieves nutrition information, and communicates with the database.
- Frontend Module: A user-friendly interface for uploading inputs, displaying nutrient results, and viewing diet history.
- Database Module: Stores user profiles, food logs, and nutritional records for personalized tracking.

3.2 Functional Requirements

The functional requirements of NutriVision define what the system must do to achieve its objectives:
1. User Registration & Login – Secure authentication and user profile management.
2. Food Image Upload – Allows users to upload images of meals for automatic recognition.
3. Speech Input Support – Converts audio meal descriptions into text for processing.
4. Food Recognition – Classifies food using trained CNN models.
5. Nutrient Estimation – Retrieves calorie and nutrient data from FoodData Central.

6. Result Display – Shows detailed nutritional breakdown.
7. History Tracking – Maintains a log of analyzed meals.
8. Search and Filter – Allows searching and filtering food logs.
9. Data Management – Admin monitoring and dataset updates.

3.3 Non-Functional Requirements

1. Performance – System should respond within 3–5 seconds.
2. Scalability – Should handle multiple users concurrently.
3. Reliability – Maintain 90% classification accuracy.
4. Usability – Responsive and intuitive UI.
5. Security – Secure authentication and encrypted communication.
6. Maintainability – Modular and updatable system design.
7. Availability – 99% uptime on cloud deployment.
8. Portability – Cross-platform compatibility.

3.4 Hardware & Software Requirements

Hardware Requirements:
- Processor: Intel i5 or higher
- RAM: Minimum 8 GB
- Storage: Minimum 100 GB HDD/SSD
- Internet Connection: Required for API access
- GPU: Optional for model training

Software Requirements:
- OS: Windows 10 / Linux / macOS
- Backend: Flask / Django / Node.js
- Frontend: React.js / HTML / CSS / Tailwind CSS
- Database: MySQL / MongoDB
- AI Libraries: TensorFlow / PyTorch
- Speech Processing: OpenAI Whisper
- IDE: Visual Studio Code / PyCharm
- Cloud: AWS / Render / Google Cloud

3.5 Technology and Tools Utilized

- React.js – Web UI development.
- Tailwind CSS – Styling and responsive design.
- Flask/Django – Backend API and AI model integration.
- TensorFlow/PyTorch – Food classification models.
- OpenAI Whisper – Speech-to-text processing.
- MySQL – Database management.
- FoodData Central API – Nutritional data source.
- GitHub – Version control.
- AWS/Render – Cloud deployment.

3.6 Constraints of Working

1. Dataset Limitations – Regional food diversity can affect accuracy.
2. Lighting and Image Quality – Poor visuals may cause misclassification.
3. Speech Recognition Errors – Accents or noise may cause transcription errors.
4. API Dependence – Third-party APIs may face rate limits or downtime.
5. Processing Delay – High-resolution images may cause latency.
6. Storage Constraints – Continuous logging increases storage demands.
7. Network Dependency – Requires an active internet connection.